

Python: exercises on recursion

This notebook was generated in student mode.

Exercise 1: Sum of Numbers (Recursive)

Exercise Description

Write a recursive function to calculate the sum of all integers from 1 to n.

Input Example: n = 5

Output Example: 15

The function should use recursion with a base case and recursive case to solve the problem.

Your solution

```
# Write your recursive function here  
def sum_recursive(n):  
    pass
```

Test your solution

```
# Test the function with the provided example
assert sum_recursive(5) == 15, "sum_recursive(5) should return 15"

# Test with other values
assert sum_recursive(0) == 0, "sum_recursive(0) should return 0"
assert sum_recursive(1) == 1, "sum_recursive(1) should return 1"
assert sum_recursive(3) == 6, "sum_recursive(3) should return 6"
```

Test your solution

```
# Test with larger values
assert sum_recursive(10) == 55, "sum_recursive(10) should return 55"
assert sum_recursive(4) == 10, "sum_recursive(4) should return 10"
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.'
```

Exercise 2: Factorial (Recursive)

Exercise Description

Calculate the factorial of a given number n using recursion.

Input Example: $n = 4$

Output Example: 24

The factorial of n (written as $n!$) is the product of all positive integers less than or equal to n . For example, $4! = 4 \times 3 \times 2 \times 1 = 24$.

Your solution

```
# Write your recursive function here  
def factorial(n):  
    pass
```

Test your solution

```
# Test the function with the provided example  
assert factorial(4) == 24, "factorial(4) should return 24"  
  
# Test with other values  
assert factorial(0) == 1, "factorial(0) should return 1"
```

```
assert factorial(1) == 1, "factorial(1) should return 1"
assert factorial(3) == 6, "factorial(3) should return 6"
```

Test your solution

```
# Test with larger values
assert factorial(5) == 120, "factorial(5) should return 120"
assert factorial(2) == 2, "factorial(2) should return 2"
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.'
```

Exercise 3: Fibonacci Sequence (Recursive)

Exercise Description

Generate the n-th Fibonacci number using recursion.

Input Example: n = 6

Output Example: 8

The Fibonacci sequence is: 0, 1, 1, 2, 3, 5, 8, 13, ... where each number is the sum of the two preceding ones.

Your solution

```
# Write your recursive function here  
def fibonacci(n):  
    pass
```

Test your solution

```
# Test the function with the provided example  
assert fibonacci(6) == 8, "fibonacci(6) should return 8"  
  
# Test with base cases  
assert fibonacci(0) == 0, "fibonacci(0) should return 0"  
assert fibonacci(1) == 1, "fibonacci(1) should return 1"  
assert fibonacci(2) == 1, "fibonacci(2) should return 1"
```

Test your solution

```
# Test with other values  
assert fibonacci(3) == 2, "fibonacci(3) should return 2"  
assert fibonacci(4) == 3, "fibonacci(4) should return 3"  
assert fibonacci(5) == 5, "fibonacci(5) should return 5"  
assert fibonacci(7) == 13, "fibonacci(7) should return 13"
```

Debug your solution

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```
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```

Exercise 4: Power of a Number (Recursive)

Exercise Description

Write a recursive function to calculate a^b (a raised to the power of b).

Input Example: $a = 2, b = 3$

Output Example: 8

The function should use recursion to compute the power operation.

Your solution

```
# Write your recursive function here  
def power(a, b):  
    pass
```

Test your solution

```
# Test the function with the provided example
assert power(2, 3) == 8, "power(2, 3) should return 8"

# Test with base case
assert power(5, 0) == 1, "power(5, 0) should return 1"
assert power(2, 1) == 2, "power(2, 1) should return 2"
```

Test your solution

```
# Test with other values
assert power(3, 2) == 9, "power(3, 2) should return 9"
assert power(4, 2) == 16, "power(4, 2) should return 16"
assert power(2, 4) == 16, "power(2, 4) should return 16"
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.'
```

Exercise 5: Reverse a String (Recursive)

Exercise Description

Reverse a given string using recursion.

Input Example: s = "hello"

Output Example: "olleh"

The function should use recursion to reverse the string character by character.

Your solution

```
# Write your recursive function here  
def reverse_string(s):  
    pass
```

Test your solution

```
# Test the function with the provided example
assert reverse_string("hello") == "olleh", "reverse_string('hello') should return 'olleh'"

# Test with base case
assert reverse_string("") == "", "reverse_string('') should return ''"
assert reverse_string("a") == "a", "reverse_string('a') should return 'a'"
```

Test your solution

```
# Test with other values
assert reverse_string("abc") == "cba", "reverse_string('abc') should return 'cba'"
assert reverse_string("python") == "nohtyp", "reverse_string('python') should return 'nohtyp'"
assert reverse_string("12345") == "54321", "reverse_string('12345') should return '54321'"
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.'
```

Exercise 6: Check Palindrome (Recursive)

Exercise Description

Check if a string is a palindrome using recursion.

Input Example: s = "madam"

Output Example: True

A palindrome is a word that reads the same forwards and backwards.

Your solution

```
# Write your recursive function here  
def is_palindrome(s):  
    pass
```

Test your solution

```
# Test the function with the provided example  
assert is_palindrome("madam") == True, "is_palindrome('madam') should return True"  
  
# Test with base cases  
assert is_palindrome("") == True, "is_palindrome('') should return True"  
assert is_palindrome("a") == True, "is_palindrome('a') should return True"
```

Test your solution

```
# Test with other values
assert is_palindrome("racecar") == True, "is_palindrome('racecar') should return True"
assert is_palindrome("hello") == False, "is_palindrome('hello') should return False"
assert is_palindrome("level") == True, "is_palindrome('level') should return True"
assert is_palindrome("python") == False, "is_palindrome('python') should return False"
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.'
```

Exercise 7: Count Digits (Recursive)

Exercise Description

Count the number of digits in a positive integer using recursion.

Input Example: $n = 12345$

Output Example: 5

The function should use recursion to count digits by repeatedly dividing by 10.

Your solution

```
# Write your recursive function here
def count_digits(n):
    pass
```

Test your solution

```
# Test the function with the provided example
assert count_digits(12345) == 5, "count_digits(12345) should return 5"

# Test with base case
assert count_digits(5) == 1, "count_digits(5) should return 1"
assert count_digits(9) == 1, "count_digits(9) should return 1"
```

Test your solution

```
# Test with other values
assert count_digits(123) == 3, "count_digits(123) should return 3"
assert count_digits(1000) == 4, "count_digits(1000) should return 4"
assert count_digits(99) == 2, "count_digits(99) should return 2"
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.'
```

Exercise 8: Greatest Common Divisor (Recursive)

Exercise Description

Find the greatest common divisor (GCD) of two numbers using recursion and the Euclidean algorithm.

Input Example: $a = 48, b = 18$

Output Example: 6

The Euclidean algorithm states that $\text{GCD}(a, b) = \text{GCD}(b, a \% b)$ until b becomes 0.

Your solution

```
# Write your recursive function here
def gcd(a, b):
    pass
```

Test your solution

```
# Test the function with the provided example
assert gcd(48, 18) == 6, "gcd(48, 18) should return 6"

# Test with base case
```

```
assert gcd(5, 0) == 5, "gcd(5, 0) should return 5"
assert gcd(7, 1) == 1, "gcd(7, 1) should return 1"
```

Test your solution

```
# Test with other values
assert gcd(12, 8) == 4, "gcd(12, 8) should return 4"
assert gcd(15, 25) == 5, "gcd(15, 25) should return 5"
assert gcd(17, 13) == 1, "gcd(17, 13) should return 1"
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.'
```

Exercise 9: Binary Representation (Recursive)

Exercise Description

Convert a decimal number to its binary representation using recursion.

Input Example: $n = 10$

Output Example: "1010"

The function should use recursion to convert the number by repeatedly dividing by 2 and collecting remainders.

Your solution

```
# Write your recursive function here
def decimal_to_binary(n):
    pass
```

Test your solution

```
# Test the function with the provided example
assert decimal_to_binary(10) == "1010", "decimal_to_binary(10) should return '1010'"

# Test with base cases
assert decimal_to_binary(0) == "0", "decimal_to_binary(0) should return '0'"
assert decimal_to_binary(1) == "1", "decimal_to_binary(1) should return '1'"
```

Test your solution

```
# Test with other values
assert decimal_to_binary(5) == "101", "decimal_to_binary(5) should return '101'"
assert decimal_to_binary(8) == "1000", "decimal_to_binary(8) should return '1000'"
assert decimal_to_binary(15) == "1111", "decimal_to_binary(15) should return '1111'"
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.'
```

Exercise 10: Sum of Digits (Recursive)

Exercise Description

Calculate the sum of all digits in a positive integer using recursion.

Input Example: n = 123

Output Example: 6

The function should use recursion to sum digits by repeatedly dividing by 10 and adding remainders.

Your solution

```
# Write your recursive function here
def sum_of_digits(n):
    pass
```

Test your solution

```
# Test the function with the provided example
assert sum_of_digits(123) == 6, "sum_of_digits(123) should return 6"

# Test with base case
assert sum_of_digits(5) == 5, "sum_of_digits(5) should return 5"
assert sum_of_digits(9) == 9, "sum_of_digits(9) should return 9"
```

Test your solution

```
# Test with other values
assert sum_of_digits(456) == 15, "sum_of_digits(456) should return 15"
assert sum_of_digits(1000) == 1, "sum_of_digits(1000) should return 1"
assert sum_of_digits(99) == 18, "sum_of_digits(99) should return 18"
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.'
```

Exercise 11: Sum of List Elements (Recursive)

Exercise Description

Calculate the sum of all elements in a list using recursion.

Input Example: `lst = [1, 2, 3, 4, 5]`

Output Example: 15

The function should use recursion to sum elements by processing the first element and recursively summing the rest.

Your solution

```
# Write your recursive function here
def sum_list(lst):
    pass
```

Test your solution

```
# Test the function with the provided example
assert sum_list([1, 2, 3, 4, 5]) == 15, "sum_list([1, 2, 3, 4, 5]) should return 15"

# Test with base case
```

```
assert sum_list([]) == 0, "sum_list([]) should return 0"
assert sum_list([5]) == 5, "sum_list([5]) should return 5"
```

Test your solution

```
# Test with other values
assert sum_list([10, 20, 30]) == 60, "sum_list([10, 20, 30]) should return 60"
assert sum_list([1, -1, 2, -2]) == 0, "sum_list([1, -1, 2, -2]) should return 0"
assert sum_list([7, 8, 9]) == 24, "sum_list([7, 8, 9]) should return 24"
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.'
```

Exercise 12: Collatz Conjecture Steps (Recursive)

Exercise Description

Count the number of steps to reach 1 in the Collatz sequence using recursion.

Input Example: $n = 3$

Output Example: 7

The Collatz sequence: if n is even, divide by 2; if n is odd, multiply by 3 and add 1. Continue until reaching 1.

Your solution

```
# Write your recursive function here
def collatz_steps(n):
    pass
```

Test your solution

```
# Test the function with the provided example
assert collatz_steps(3) == 7, "collatz_steps(3) should return 7"

# Test with base case
assert collatz_steps(1) == 0, "collatz_steps(1) should return 0"
assert collatz_steps(2) == 1, "collatz_steps(2) should return 1"
```

Test your solution

```
# Test with other values
assert collatz_steps(4) == 2, "collatz_steps(4) should return 2"
assert collatz_steps(5) == 5, "collatz_steps(5) should return 5"
assert collatz_steps(6) == 8, "collatz_steps(6) should return 8"
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.'
```

Exercise 13: Check if List is Sorted (Recursive)

Exercise Description

Check if a list is sorted in ascending order using recursion.

Input Example: `lst = [1, 2, 3, 4, 5]`

Output Example: `True`

The function should use recursion to check if each element is less than or equal to the next element.

Your solution

```
# Write your recursive function here
def is_sorted(lst):
    pass
```

Test your solution

```
# Test the function with the provided example
assert is_sorted([1, 2, 3, 4, 5]) == True, "is_sorted([1, 2, 3, 4, 5]) should return True"

# Test with base cases
assert is_sorted([]) == True, "is_sorted([]) should return True"
assert is_sorted([5]) == True, "is_sorted([5]) should return True"
```

Test your solution

```
# Test with other values
assert is_sorted([1, 3, 2, 4]) == False, "is_sorted([1, 3, 2, 4]) should return False"
assert is_sorted([1, 1, 2, 3]) == True, "is_sorted([1, 1, 2, 3]) should return True"
assert is_sorted([5, 4, 3, 2, 1]) == False, "is_sorted([5, 4, 3, 2, 1]) should return False"
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.'
```

Exercise 14: Find Maximum in List (Recursive)

Exercise Description

Find the maximum element in a list using recursion.

Input Example: lst = [3, 1, 4, 1, 5, 9, 2]

Output Example: 9

The function should use recursion to compare the first element with the maximum of the rest of the list.

Your solution

```
# Write your recursive function here  
def find_max(lst):  
    pass
```

Test your solution

```
# Test the function with the provided example
assert find_max([3, 1, 4, 1, 5, 9, 2]) == 9, "find_max([3, 1, 4, 1, 5, 9, 2]) should return 9"

# Test with base case
assert find_max([5]) == 5, "find_max([5]) should return 5"
assert find_max([10]) == 10, "find_max([10]) should return 10"
```

Test your solution

```
# Test with other values
assert find_max([1, 2, 3]) == 3, "find_max([1, 2, 3]) should return 3"
assert find_max([10, 5, 8, 3]) == 10, "find_max([10, 5, 8, 3]) should return 10"
assert find_max([-1, -5, -3]) == -1, "find_max([-1, -5, -3]) should return -1"
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.'
```

Exercise 15: Count Zeros in Number (Recursive)

Exercise Description

Count the number of zeros in a positive integer using recursion.

Input Example: n = 10203

Output Example: 2

The function should use recursion to count zeros by examining each digit.

Your solution

```
# Write your recursive function here  
def count_zeros(n):  
    pass
```

Test your solution

```
# Test the function with the provided example  
assert count_zeros(10203) == 2, "count_zeros(10203) should return 2"  
  
# Test with base cases  
assert count_zeros(0) == 1, "count_zeros(0) should return 1"  
assert count_zeros(5) == 0, "count_zeros(5) should return 0"
```

Test your solution

```
# Test with other values
assert count_zeros(1000) == 3, "count_zeros(1000) should return 3"
assert count_zeros(123) == 0, "count_zeros(123) should return 0"
assert count_zeros(50607) == 2, "count_zeros(50607) should return 2"
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.'
```

Exercise 16: Flatten Nested List (Recursive)

Exercise Description

Flatten a nested list structure using recursion.

Input Example: `nested_list = [1, [2, 3], [4, [5, 6]], 7]`

Output Example: `[1, 2, 3, 4, 5, 6, 7]`

The function should use recursion to handle arbitrarily nested lists and return a flat list.

